Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Solid-state hydrogen storage for heavy-duty applications*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		*Energy & Resources*	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / Rail Vehicles	*Low-Emission and Alternative-Energy Vehicle Technologies*	*New & Renewable Energy*	*Hydrogen*
Concept of Proposed Technology	*The proposed R&D project aims to develop and demonstrate a solid-state hydrogen energy system for heavy duty sector that can be integrated into a small heavy-duty vehicle, such as a forklift, achieving near-market readiness. The core concept is to retrofit an existing industrial vehicle platform with a metal-hydride-based hydrogen storage system combined with a fuel-cell power system, delivering zero-emission electric propulsion without requiring major structural modifications. The solution is designed for indoor and industrial use, with a strong focus on compactness, intrinsic safety, and ease of integration into existing vehicle fleets.* *The solid-state hydrogen storage system will be based on metal hydrides, enabling low-pressure hydrogen storage with high volumetric energy density. Compared to compressed hydrogen systems,*			

	metal hydrides significantly reduce risks related to leakage and explosion, making them particularly suitable for industrial environments subject to strict workplace safety requirements. The high mass of the metal-hydride storage system will be deliberately utilized as a functional counterweight, especially for forklift applications, improving vehicle stability while eliminating the need for separate ballast components. *Depending on the Korean SME core expertise the project can integrate hydrogen storage, thermal management, fuel-cell power generation, power electronics, and/or vehicle control into a unified system. When aimed, hydrogen refueling can be designed for low-pressure operation, targeting storage of hydrogen sufficient for one full workday and refueling times of approximately 30 minutes to reach 80 percent storage capacity. By the end of the project, a fully operational prototype hydrogen storage unit or vehicle can be demonstrated under real industrial operating conditions, achieving TRL 6-7 and providing a clear pathway toward commercialization and production ramp-up.*
Rationale for Joint R&D Project	*Small heavy-duty vehicles such as forklifts play a critical role in industrial logistics and are increasingly targeted for decarbonization under Korean and international climate and hydrogen economy strategies. Current zero-emission solutions are dominated by battery-electric and compressed-hydrogen fuel-cell systems, both of which face limitations in industrial use. Battery systems suffer from limited operating time and charging downtime, while compressed hydrogen raises safety, space, and regulatory concerns, particularly for indoor applications.* *Solid-state hydrogen storage based on metal hydrides offers a promising alternative by enabling low-pressure, intrinsically safe hydrogen storage with high volumetric density. In addition, the ability to use the storage system as a structural counterweight provides a unique advantage for forklifts and similar vehicles. Despite this potential, solid-state hydrogen storage for heavy-duty vehicle applications remains at an early demonstration stage and requires coordinated R&D at the system and vehicle-integration level.* *A joint R&D project between Fraunhofer IFAM and a Korean SME enables the combination of advanced German expertise and IP in metal-hydride materials (1-3), thermal management, and system integration with Korean strengths in vehicle engineering, manufacturing, regulation, and market access. Public support is essential to de-risk this high-impact but technically challenging innovation, bridge the gap to TRL 7 demonstration, and align development with Korean industrial, regulatory, and hydrogen mobility roadmaps*
Key Words	*1. Heavy duty* *2. Solid state hydrogen storage* *3. Low pressure* *4. Intrinsic safety* *5. Hydrogen*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4-5)	End of the Project	Level (6-7)
R&D Objectives	*Main objective of the project is to develop and demonstrate a solid-state hydrogen storage unit for heavy duty applications coupled to fuel-cell power system that can be integrated (also within a project) into a small heavy-duty vehicle. In the best case, the prototype should be validated under real industrial operating conditions. The project aims to deliver a retrofit-capable, low-pressure hydrogen energy system that meets industrial performance, safety, and regulatory requirements while enabling zero-emission operation.* *Key objectives include achieving hydrogen storage sufficient for a full work shift, fast refueling comparable to conventional industrial vehicles, and reliable power delivery under typical duty cycles. The project also aims to demonstrate the dual use of metal-hydride storage as both energy carrier and vehicle counterweight, as well as long-term durability and material cycling capability. The final outcome might include a validated vehicle prototype, system specifications, and a commercialization and production ramp-up plan supporting TRL 7 deployment.*			
R&D Target Specifications	*The solid-state hydrogen storage system will target low-pressure operation below 30 bar, with a high volumetric hydrogen storage density suitable for compact vehicle integration. The system shall store sufficient hydrogen to support one full industrial workday, depending on vehicle class and duty cycle, and enable refueling of at least 80 percent capacity within approximately 30 minutes using low-pressure hydrogen infrastructure.* *The integrated fuel-cell power system shall provide continuous, CO-free electric power meeting the performance requirements of small heavy-duty vehicles, including peak and steady-state power demands. The metal-hydride storage system shall demonstrate stable cycling over several thousand hydrogen absorption and desorption cycles with limited capacity degradation. From a safety perspective, the system shall comply with Korean hydrogen safety and industrial vehicle regulations, incorporating thermal management, hydrogen leak detection, and fail-safe design. The final vehicle demonstrator shall achieve TRL 7, validated in real industrial environments.*			
R&D Plan	*Overall approach and timeline (36 months total)* *The project is structured into two sequential phases, progressing from market analysis and operational definition to manufacturing of full-scale prototype and possible demonstration of retrofitted small heavy-duty vehicle.*			

	Phase I Pre-study and System Definition (6 months) • *Definition of operational requirements including power demand, duty cycles, operating time, and refueling strategy* • *Analysis of existing heavy-duty vehicle platforms and interfaces enabling retrofit without major structural modifications* • *Review of Korean hydrogen safety, industrial vehicle, and workplace regulations, including classification under the High-Pressure Gas Safety Control Act* • *Selection of suitable metal-hydride materials and preliminary system architecture* • *Initial system simulations including thermal behavior and heat exchange* • *Definition of integration concept, key parameters, and preliminary business and exploitation strategy* *Milestone Phase I: Approved vehicle and system concept with defined requirements and go decision for prototype development* ***Phase II Main R&D, Integration, and Demonstration (30 months)*** • *Development and optimization of metal-hydride storage materials and system design* • *Construction of hydrogen storage vessel and integration of thermal management, sensors, and safety systems* • *Manufacturing of full-scale prototype storage and power systems suitable for vehicle integration* • *Integration of fuel cell, power electronics, battery hybridization, and vehicle control systems* • *Laboratory and vehicle-level testing for performance, safety, and durability* • *Demonstration of the retrofitted vehicle in real industrial operating environments* • *Preparation of production ramp-up, cost-efficiency analysis, and certification documentation* *Milestone Phase II: Validated TRL 8 solid-state hydrogen-powered heavy-duty vehicle demonstrator*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Fraunhofer IFAM will provide scientific and technological co-leadership in metal-hydride storage systems, including material selection, laboratory-scale testing, and performance characterization. The institute will support system design through thermal modeling, simulation of heat exchange behavior, and experimental validation in test rigs. Fraunhofer IFAM will contribute system-level knowledge for metal-hydride storage, safety assessment, and durability testing. The institute will also support integration concepts and knowledge transfer to the industrial partner.*
Expected R&R of Korean SME Partner	*The Korean SME will lead vehicle integration, system engineering, and manufacturing activities, including construction of the hydrogen storage vessel and integration into the vehicle platform. The SME will provide expertise in real-life application requirements, Korean regulatory frameworks, and industrial vehicle standards. Responsibilities include fuel-cell system integration, power electronics, hybrid powertrain control, and battery management. The SME will also lead commercialization planning, certification preparation, and market deployment strategies.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The project will deliver up to a TRL 7 solid-state hydrogen storage and fuel-cell power system possibly integrated into a small heavy-duty vehicle, demonstrated under real industrial operating conditions. Technologically, it will establish a safe, low-pressure, and compact alternative to batteries and compressed hydrogen systems, with particular advantages for indoor and high-utilization applications. The demonstrator will provide validated evidence of endurance, safety, and zero-emission operation.* *Economically, the project is expected to reduce downtime and operating costs for industrial users while supporting the emergence of new retrofit and vehicle-supply markets. The solution is expected to capture 10–15 percent of the Korean zero-emission forklift market in the early commercialization phase, with potential expansion to international markets in Asia, Europe, and North America. The project will strengthen Korea's hydrogen mobility ecosystem, support national decarbonization goals, and create high-skilled jobs across hydrogen systems engineering, manufacturing, integration, and maintenance.*

6. References

(1) EP3045910 B1 patent, Messeinrichtung und Messverfahren zur Bestimmung der Menge eines in einem Speicher aufgenommenen Gases an einem porösen Speichermaterial

(2) EP 3 118 511 B1 patent, Speicherelement für Gase

(3) DE 10 2020 204 214 B4 patent, Behälter zum Speichern von Wasserstoff

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization